

12

Power

I'm ambivalent about statistical power. On one hand, if we're using NHST, power is a vital part of research planning. Also, funding bodies and ethics review boards often require power calculations. On the other hand, power is defined in terms of NHST, so if we don't use NHST we can ignore power and instead use precision for research planning, as I discuss in Chapter 13. However, I feel it's still necessary to understand power, partly because power calculations are often required, and partly to help understand NHST and its weaknesses. I have therefore included this chapter, although I hope that, sometime in the future, power will need only a small historical mention.

Statistical power has a narrow technical definition, but sometimes “power” is used more broadly to refer to the extent that an experiment provides information to help solve our research problems, and gives us insight about the world. I'll use the term *informativeness* to refer to this more general characteristic of an experiment, and I'll distinguish it from the less important concept of statistical power. Informativeness will remain an important idea, even if we move on from NHST and no longer use statistical power.

In this chapter we'll discuss

- An introduction to power
- A take-home image—the power picture
- Calculating power
- Intuitions about power
- Post hoc power—*illegitimate* power
- What we really want—informativeness
- High power, high informativeness
- Reporting power

Introduction to Power

Back in Chapter 2 I defined statistical power: It's the probability of obtaining statistical significance—and therefore rejecting H_0 —if a precisely

Statistical power is the probability of obtaining statistical significance if the alternative hypothesis is true, that is, if there really is a population effect of a stated size.

stated alternative hypothesis is true. It's the chance that we'll be able to reject the null hypothesis if there's a true effect of a particular stated size in the population.

Informally, power is the chance that our experiment will identify a particular real effect, if it exists.

Calculating power requires the alternative hypothesis to be a precise statement of an ES, for example, that a new analgesic reduces headache by exactly 1.0 scale units on our 0 to 10 pain scale. It's much more common for researchers to use NHST to test H_0 against a vague alternative, such as $H_1: \delta \neq 0$, and so having to choose a precise value for H_1 is an unfamiliar task. It's also a difficult task because, most likely, the more pain relief the better, and there isn't any exact amount of pain relief in which we're particularly interested. Therefore, the precise H_1 is often an arbitrary choice. In addition, it may or may not be an estimate of the population ES that's true in the world. I'll refer to it as *target* δ to emphasize that

Target δ is a precise value of the population ES that we choose for the H_1 . It may or may not be an estimate of the population ES that's true in the world.

it's our choice of a value to use in a power calculation. So target δ is a value that's specified by H_1 , and it can be any value we care to choose for our power calculation.

To illustrate some basic features of power I'll discuss the simulation we used in Chapter 5 for the dance of the p values. It was a two-independent-groups experiment that evaluated your new relaxation therapy. We assumed that $\delta = 0.5$ for your therapy, and that σ was known. We used groups of size $N = 32$. For power calculations we chose target $\delta = 0.5$, the value assumed true in the simulation. ESCI calculated that, with two-tailed $\alpha = .05$, the power is .52. Back in Chapter 5, Figure 5.11 shows the frequency histogram of the p values we obtained in 1,500 simulations of that experiment. Close to 52% of them were less than .05, which illustrates power as the chance that an experiment will reject the null hypothesis when target δ is true in the (simulated) world.

As a further illustration of power, I fired up the **Dance p** page of **ESCI chapters 5–6** and set up the simulation as before, then increased N until ESCI reported near red 3 that power is .80. Using $N = 63$ did the trick. I then ran 1,500 simulations and enjoyed the dance of the CIs and the dance of the p values. Figure 12.1, which is analogous to Figure 5.11, shows the frequency histogram of the 1,500 p values I obtained. Above each column, the observed and theoretically predicted percentages of p values in that column are displayed, where “theoretically predicted” refers to the result of a power calculation using our target $\delta = 0.5$. Adding the observed percentages for the *, **, and *** columns indicates that 80.4% of my p values were less than .05, close to the 80.1% total of the three theoretically

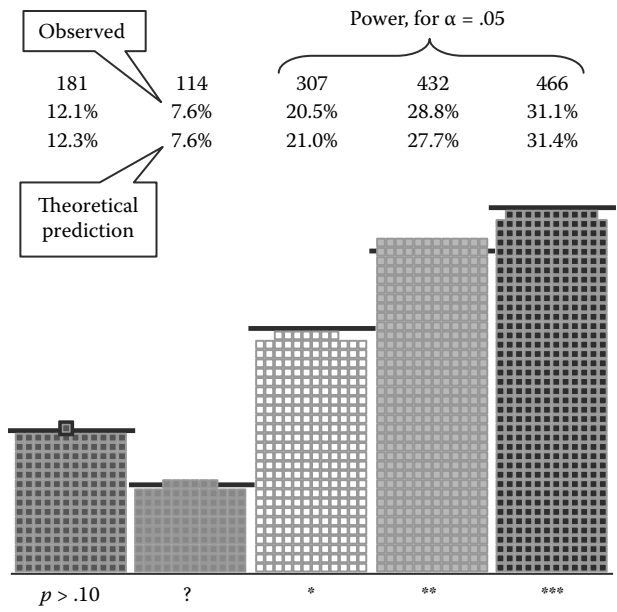


FIGURE 12.1
The frequency histogram of p values, for 1,500 simulated experiments that evaluated your new relaxation therapy. The experiments used two independent groups, each with $N = 63$, and $\delta = 0.5$, so, with two-tailed $\alpha = .05$, the power = .80. Back in Chapter 5, Figure 5.11 shows the corresponding histogram when $N = 32$ and power = .52. Above each column the top number reports the frequency for that column, then the numbers below are the observed and theoretically expected percentages of p values in that column. The sum of the percentages for the *, **, and *** columns is power.

predicted percentages—which is the power of .80. Here are my comments on this illustration of power:

- We had inside knowledge that the simulation assumed $\delta = 0.5$, and therefore we could select target $\delta = 0.5$ for our power calculation. Because those two values were the same, the percentage (80.4%) of simulated experiments that rejected H_0 matched closely our calculated power of .80. If we simulated many more experiments, the match would be even closer.
- To calculate power we need to know the experimental design, the sample size N , our α , and a target δ . We changed N from 32 to 63 and saw power change. Change any of those features of an experiment and power will change.
- It therefore makes no sense to say “my experiment has power of .85” if you don’t tell us all those other things about your experiment.

The most common omission is target δ . Any stated value of power describes a particular experiment, for a stated exact target δ .

A stated value of statistical power refers to an experiment with a specified design, for stated values of N , α , and target δ . In particular, target δ must be specified.

- The histogram in Figure 12.1 can also tell us power for some other values of two-tailed α . For example, for $\alpha = .01$ we add percentages for the ** and *** columns and find that 59.9% of my p values were less than .01, close to the 59.1% total of the two theoretically expected percentages. Power was thus .59, which is lower than .80 because we adopted a more stringent $\alpha = .01$ criterion for statistical significance. Yes, change α , and power will change.
- Power is a single value, say .80, but it's based on a distribution of p values. We found in Chapter 5, and Figure 12.1 further demonstrates, that this distribution of p is very wide. Recall how very wide p intervals usually are: For all but extremely high values of power, any experiment can give a p value from a very wide range.

A more general conclusion from all this discussion of NHST, p values, and power may be that our conclusion in Chapter 5 was correct: It's more informative to use estimation instead, and to have in mind meta-analysis to combine evidence over experiments. I hope this discussion of power may reinforce such new statistics intuitions, which are worth keeping in mind even as I continue to use NHST to discuss power.

It's time for a picture: I believe a picture is needed to make power—like just about any other concept—understandable and memorable. The power picture is my suggestion for a take-home image. I'll use that picture as a context to describe power and how to calculate it.

The Power Picture

Recall the Hot Earth Awareness Test (HEAT)—the fictitious test I introduced in Chapter 3. Suppose we're investigating HEAT scores in our country. The HEAT has population mean $\mu = 50$ and standard deviation $\sigma = 20$ in the reference population used in the development of the test, so we'll use $H_0: \mu = \mu_0 = 50$ as our null hypothesis. We'll take a sample of 10 students—way too small, I know, but small N emphasizes interesting features of the power picture. For such a single-group design, assuming σ is not known, the test statistic is

$$t = \frac{M - \mu_0}{s/\sqrt{N}} \quad (10.1)$$

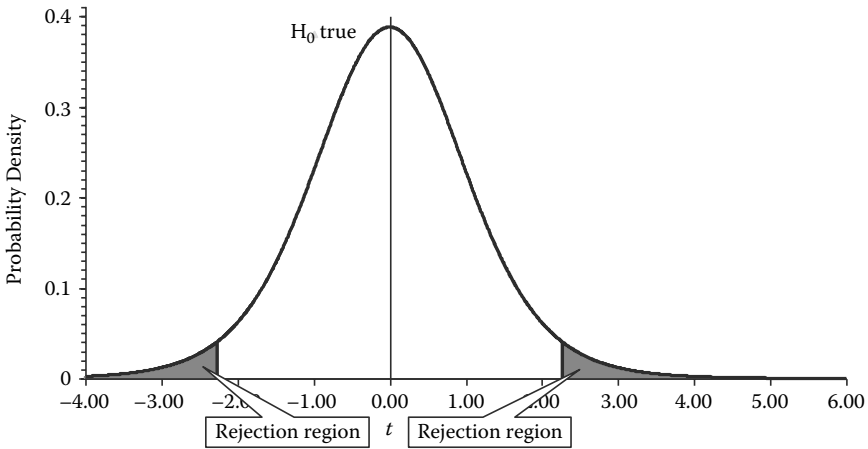


FIGURE 12.2

The distribution of t if H_0 is true, for a single group with $N = 10$, assuming σ is not known. The rejection region for a two-tailed test with $\alpha = .05$ is the two shaded tail areas, which extend above $t = 2.26$ and below $t = -2.26$. The curve is the central t distribution with $df = (N - 1) = 9$.

We encountered that formula in Chapters 10 and 11. We'll choose $\alpha = .05$ and, as usual, a two-tailed test. We can insert $\mu_0 = 50$, $N = 10$, and our sample M and s into Equation (10.1) to calculate t . Figure 12.2 shows the distribution of t if the null hypothesis is true. The rejection region for our test is the two tails of that distribution beyond plus and minus the critical value of t , which is $t_{.95}(N - 1) = t_{.95}(9) = 2.26$. We encountered $t_{.95}(N - 1)$ back in Chapter 3, Equation (3.3), when we were calculating 95% CIs. As usual, you can use the **Normal z t** page of **ESCI chapters 1–4** to find that critical value, with help from Appendix B if you wish. Figure 12.2 marks the two tails that make up the rejection region. We reject the null hypothesis if the t value we calculate falls in either tail.

A Point Alternative Hypothesis

To calculate power, we need to specify a target δ , which is the exact value stated by the alternative hypothesis. As I mentioned earlier, target δ is a value we choose that may or may not be an estimate of the true population ES. I'll discuss below how we might choose target δ . For our example here, suppose a large national survey reports that the mean HEAT scores for students in Awareland is 68. We hope students in our country are just as aware of climate change issues, so we'll choose $H_1: \mu_1 = 68$ as our point alternative hypothesis. The corresponding target population ES is then

$$\delta = \frac{\mu_1 - \mu_0}{\sigma} = \frac{68 - 50}{20} = 0.90 \tag{12.1}$$

In other words, μ_1 is 0.90 of a population SD higher than μ_0 . In terms of Cohen's conventions, that's a large effect. To find power we need to know the distribution of the t statistic, as given by Equation (10.1), when the alternative hypothesis is true. Figure 12.3 is an example of what I call the *power picture* and shows that distribution. Power is the area under that H_1 distribution for all values of t in the rejection region, which means power

The *power picture* is what I call a figure like Figure 12.3 that displays the distributions of the test statistic, z or t , when H_0 is true, and when H_1 is true.

is the large area under that curve corresponding to the right rejection tail, plus a tiny invisible area under the H_1 curve corresponding to the left rejection tail. We can usually ignore that second, tiny area,

although ESCI doesn't. The large area to the right is shaded dark gray, although note that this area includes the right tail rejection area shaded light gray—think of the dark gray continuing behind the light gray.

For our single group example, with $N = 10$, $\alpha = .05$, and target population ES of 0.90, the power is .72. The dark shaded area in Figure 12.3 is 72% of the total area under the H_1 curve. If we run our experiment, and students in our country really do have a mean HEAT score of 68, then

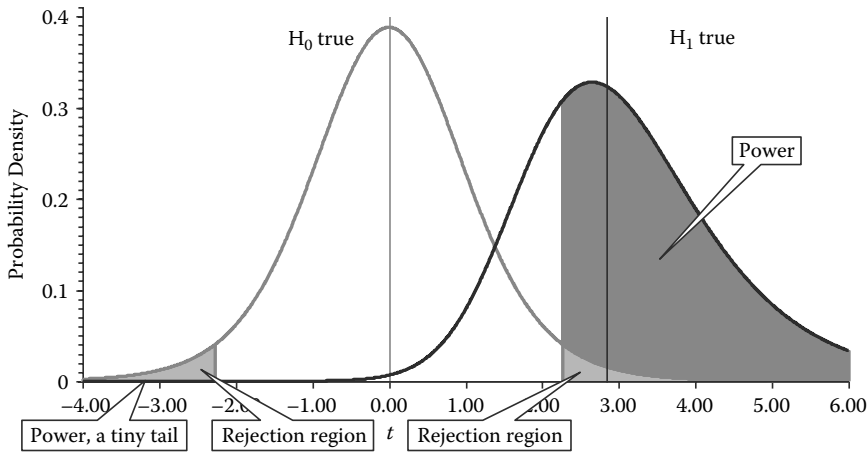


FIGURE 12.3
The example of Figure 12.2, but showing also the distribution of t when H_1 is true. The population ES under H_1 of $\delta = 0.90$ determines the H_1 distribution, which is noncentral t with noncentrality parameter $\Delta = 2.85$ and $df = 9$. The vertical line on the right marks $\Delta = 2.85$. Power = .72, which is the area under the H_1 curve for t in the rejection region. This is the area on the right corresponding to the right rejection area tail, plus a tiny invisible area under the H_1 curve corresponding to the left rejection area tail.

there's a 72% chance that we'll obtain $p < .05$ and can reject the null hypothesis that their population mean is 50. Even with a large population ES we have a $100 - 72 = 28\%$ chance of failure, because our sample size is so small.

You may feel that my example is contrived and unrealistic, and the choice of an exact value for the alternative hypothesis may seem especially artificial. Why not simply use our M and its CI to give an estimated ES for students in our country? The CI would, no doubt, be wide and tell us that we have only an imprecise estimate, but that's the truth. Yes, such doubts about these power discussions are justified, and it's worth keeping in mind the estimation alternative as we continue to discuss NHST.

Sampling Distribution When H_1 Is True

It's easiest to think of power in terms of the power picture and areas under the H_1 curve. How would Figure 12.3 change for a different population ES under H_1 , meaning a different value of μ_1 and therefore a different value of target δ ? The H_1 curve would shift—to the right for larger δ and to the left for smaller δ . Figure 12.4 is another version of the power picture that illustrates a smaller δ and an H_1 curve shifted left. Less of the curve corresponds to the right rejection tail, so power is smaller. That's what we'd expect: A smaller true effect is less likely to give $p < .05$. In Figure 12.4 I'm using $\mu_1 = 60$, and so $\delta = 0.50$, which gives power of .29.

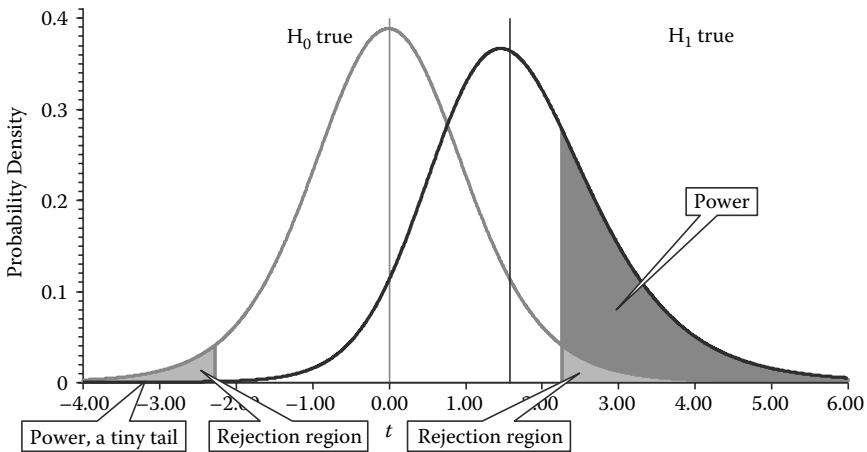


FIGURE 12.4

Same as Figure 12.3, but with population ES of $\delta = 0.50$, so the H_1 curve is shifted left. That curve is a noncentral t distribution with noncentrality parameter $\Delta = 1.58$ and $df = 9$. The vertical line on the right is positioned at $\Delta = 1.58$. Power is now .29.

The H_1 curve is a noncentral t distribution. If that's sufficient (or too much) information about the H_1 distribution, feel free to skip to the next paragraph. To find the noncentrality parameter Δ , recall from Chapter 10 that

$$\Delta = \frac{\mu_1 - \mu_0}{\sigma/\sqrt{N}} \quad (10.3)$$

When σ is unknown, t is our test statistic. When the alternative hypothesis is true, t has a noncentral t distribution, so power is an area under a noncentral t curve.

Substitute from Equation (12.1) to obtain the beautiful relation that

$$\Delta = \delta\sqrt{N} \quad (12.2)$$

The noncentrality parameter Δ simply reflects the target population ES and the sample size. For Figure 12.3, $\delta = 0.90$ and Equation (12.2) gives $\Delta = 0.90 \times \sqrt{10} = 2.85$. This value is marked by the vertical line at approximately the mean of the H_1 curve. Similarly, for Figure 12.4, $\delta = 0.50$ and $\Delta = 0.50 \times \sqrt{10} = 1.58$, and this is the value marked by the vertical line. In any such case, to find power we need to calculate an area under the noncentral t distribution.

This would be a good moment to cross-check with some other statistics textbook you know. If it discusses power, it probably has a figure like Figure 12.3, but note two likely differences. First, it probably assumes that σ is known and therefore shows two normal distributions. That's a perfectly fine way to introduce power, and is the approach I used for the dance of the p values in Chapter 5. It's more realistic, however, to assume that σ is not known, in which case we need the central and noncentral t distributions, as in Figures 12.2 to 12.4.

Second, in Figures 12.3 and 12.4 we're assuming that H_1 is true, so I faded the H_0 curve and the rejection regions to light gray, whereas I highlighted the H_1 curves in black and areas for power in dark gray. Most textbooks simply show two similar curves in their power picture. That may be misleading because it doesn't emphasize that at any moment only one hypothesis can be true. I used the strategy of highlighting and graying out in Chapter 10 when discussing sampling, first assuming H_0 true, then assuming H_1 true. When examining any power picture, it's essential to be clear which hypothesis you are currently assuming true, and it may help if the picture itself makes that obvious.

ESCI's Power Picture

Figures 12.2 to 12.4 come from the **Power picture** page of **ESCI chapters 10–13**. If you fire up that page, you can explore many things about power

esci **Power picture**
 See statistical power for a single sample
 If σ known, use Z, see normal distributions
 If σ not known, use t , see t distributions

1 Set up
 Sample size
 N 10
 df 9
☐ Use Z, assume σ known
☒ Use t , assume σ not known
☒ Display H_0 distribution
☒ Display line at H_0 mean

2 Effect size Population effect size if H_1 true
 δ 0.90
☒ Display H_1 distribution
☒ Display line for H_1
 Display noncentral t curve? Yes Δ 2.846

3 Hypothesis test
 α 0.05
 Assume true
☐ H_0 ☒ H_1
☒ Show rejection region
☒ Rejection region to front
 Critical value of t 2.262

4 Power
☒ Show power
Power 0.717

5 << Shift t axis >> In Zoom t axis Out

FIGURE 12.5

The control area in the **Power picture** page of **ESCI chapters 10–13**.

for the single-group case. Figure 12.5 shows the control panels; power is reported near red 4. Here are some suggestions:

- Use the spinner near red 1 to select sample size N . Change N freely, because intuitions about sample size are central to understanding power.
- Click the radio buttons near red 3 to specify whether you are assuming that H_0 or H_1 is true. See corresponding aspects of the figure highlighted or faded to gray.
- Click the radio buttons near red 1 to specify whether or not σ is assumed known. Note the change between using z as the test

statistic and seeing two normal curves, and using t and seeing central and noncentral t curves.

- Use the slider near red 2 to change target δ , the population ES if H_1 is true. The H_1 curve shifts left and right: Watch the big changes in its shape and in power, which is reported near red 4.
- Use the spinner near red 3 to change two-tailed α .
- Use this page to calculate power for a single group experiment.

Calculating Power

A Two-Independent-Groups Example

Suppose you plan with a colleague in another country to compare HEAT scores. She can organize a sample of 80 students, and you can find 60. You decide to use $\alpha = .05$, and agree that the difference in mean HEAT scores between countries may only be small, say $\delta = 0.2$, so that's your target δ .

To calculate power for the two-independent-groups case, fire up the **Simulate two** page of **ESCI chapters 5–6**, scroll far right to see checkboxes at red 11 and 12, click them both on, then scroll back left to see a blue panel for power at red 13. Figure 12.6 shows part of the control area for that page, including the panel for power.

Set the two sample sizes at red 3, and use the spinner for C near red 6 to adjust the α value shown at red 13. Adjust the spinners near red 7 for population parameters μ_1 , μ_2 , and σ until $\delta = 0.2$ is shown in the red 7 area. You can choose any convenient values that give $\delta = 0.2$, which is what's used to calculate power. For example, set $\mu_1 = 50$ and $\sigma = 10$, then adjust μ_2 until you see the δ you want. Click **Another experiment** near red 1 to trigger calculation of d , and power using t . That's how I created the screen shot shown in Figure 12.6. Power is .21 assuming that you'll use t as the test statistic. (Using z it's .22—very similar, as we might expect given the fairly large samples.) Oh dear, such low power! You investigate other options. Increasing both sample sizes to 100 only increases power to .29. However, you find that the seemingly small change of target δ from 0.2 to 0.3 (as before, adjust μ_2 to obtain the δ you want) lifts power to .56. That's better, but hardly good. Hmm, can you just change target δ at whim like that? Later I'll discuss ways to choose and justify target δ , but for the moment we'll assume that it's simply your choice. Using $\delta = 0.4$ gives power .80, and $\delta = 0.5$ gives power .94. Much better. You know that you can also increase power by increasing α , but you agree with your colleague that increasing α is a bad strategy. You recall the conclusion of Chapter 5 that

The screenshot shows a software interface with three main sections:

- 6 Confidence Intervals**: Includes a checked box for "Display CIs", a "C" value of 95, and an unchecked box for "Display CI overlap".
- 7 Two populations**: Includes a checked box for "Show values of population parameters", mean values $\mu_1 = 50$ and $\mu_2 = 52$, a standard deviation (SD) $\sigma = 10$, an effect size $\delta = 0.2$, and an unchecked box for "Display population means $\pm$ SD".
- 13 Power**: Shows a two-tail α of .05. It compares two methods: "Using σ and z " with a power of .22, and "Don't use σ , so use t " with a power of .21. There is an unchecked box for "Show post hoc power" and a corresponding empty box.

FIGURE 12.6

Part of the control area in the **Simulate two** page of ESCI chapters 5–6. Sample sizes are $N_1 = 80$ and $N_2 = 60$.

only very small p values tell us much at all, so you'd strongly prefer to use $\alpha = .01$. You set $C = 99$ at red 6, so $\alpha = .01$, then find that, for $\delta = 0.2$, power is a pathetic .12. For target $\delta = 0.50$, however, power is a more encouraging .82. You agree to hope for the best and go ahead with the experiment using $\alpha = .01$ and two groups of 100.

My imaginary discussion illustrates a common way to use power calculations to guide research planning, by exploring trade-offs between various features that determine power. That's useful because it helps us understand our options, but, on the other hand, it offers scope for fudging: Set δ just a little higher and we'll get a more respectably high value for power. I'll discuss that fudging problem later. First, however, note that the calculations are based on the assumptions we usually make, but which may or may not be reasonable in a particular real-life situation. We're assuming that each sample is a random sample from a normal population, and that the two populations—students' HEAT scores in the two countries—have the same SD. Using those assumptions, we've found that power is often disappointingly low, that larger sample sizes can increase power but only to a moderate extent, and that to use a more reassuring $\alpha = .01$, rather than .05, we pay a considerable price in terms of

In general, power is influenced by sample size, but is influenced more by α and even more by δ , the target population ES.

decreased power. In addition, target δ , the population ES we're aiming to find, has a dramatic influence on power. All of those

conclusions are important and quite general facts about experimental design that are highlighted by power calculations. These harsh realities about the difficulty of designing experiments with high power may be a big reason why researchers usually don't calculate or report power. Perhaps they really don't want to know how low it is?

Finally, for this two-independent-groups example, consider the experiment you decided to run: Two groups of 100 seems, by the standards of many research fields, quite comfortable. But even with those groups, it takes a medium-sized effect of $\delta = 0.5$ before power reaches a comfortable .94 using $\alpha = .05$, or .82 using $\alpha = .01$. Cohen (1969, 1988) introduced the custom that power of .8 may be acceptable. Medicine also sometimes uses .8 as a planning target for power, although .9 or even higher values are sometimes used. However, if we go with the custom and use power of .8, there's still only an 80% chance of obtaining $p < \alpha$ if our target δ is a true effect. Running the experiment amounts to choosing a p value from the wide distribution in Figure 12.1, and a 20% chance that we'll be disappointed. Power of .8 may be hard to achieve, but it is hardly high enough for comfort. If you run your HEAT study with two samples of 100 and $\alpha = .01$, then power is .82 and you have an 18% chance of failing to obtain statistical significance even if the true effect is $\delta = 0.5$. Perhaps the slogan for this chapter should be, "No free lunch!?" Meta-analysis of a number of experiments springs to mind: Yes, indeed.

A Paired Design Example

A further and important way to improve things is to use a more sensitive experimental design. Suppose you are interested in possible changes in HEAT scores in your country over the last 2 years. You could compare two independent groups of students at the two times, but, fortunately, 2 years ago a sample of 40 students gave HEAT scores and you are able to test the same students again this year. You decide that the paired design is appropriate, so you investigate the **Simulate paired** page of **ESCI chapters 5–6**. Scroll far right, click at red 13 and 14 then scroll back left to see the blue power panel at red 15. Figure 12.7 shows part of the control area, including the power panel. Set N near red 4, and α by adjusting C near red 7. For the paired design you also need to set a value for the correlation in the population between the two measures. You decide to use .9, because you expect there will be a very high correlation between HEAT scores at the two times. You need to adjust both σ , the SD within each population, and σ_{diff} , the SD of the population of

7 Confidence Intervals

☒ Display CI on measures C 99

☒ Display CI on differences

8 Populations of measures, and differences

☒ Show values of population parameters

Means μ_1 10 μ_2 18

SD σ 40 σ_{diff} 18

Correlation .9 Effect size δ 0.2

☐ Display population means $\pm$ SD

15 Power

Two-tail α .01 Power

Using σ and Z .59

Don't use σ , so use t .55

☐ Show post hoc power

FIGURE 12.7

Part of the control area in the **Simulate paired** page of **ESCI chapters 5–6**. Sample size is $N = 40$.

paired differences, which together determine the population correlation between the two measures. Recall from Chapter 6 that the higher this correlation, the more sensitive the design. Here's an easy way to do it: Set $\mu_1 = 10$ and $\sigma = 40$, then adjust μ_2 until δ is as you want it, for example, $\delta = 0.2$. Now adjust σ_{diff} until the correlation shown near red 8 is as close as possible to what you want, for example, correlation = .9. Click **Another experiment** to trigger calculation, and see the power values near red 15.

You play around with various values and find that, when you set $N = 40$, correlation of .9, and $\alpha = .01$, the power is .55 for $\delta = 0.2$, as Figure 12.7 illustrates. For $\delta = 0.3$ power is a wonderful .93. Even with correlation of .8, power values are considerably higher than for a comparable two-independent-groups experiment. Once again you find that N makes a difference, but it's target δ and the correlation between the two measures that have the really big influences on power.

Yes, the paired design can be powerful and, as we saw in Chapter 6, it's an attractive design, but, of course, only when it suits the situation.

Might carry-over effects between the two measures be a problem? As ever, careful thinking is needed. In this case, be very clear about the two ways of investigating the question, “Have HEAT scores changed over the last 2 years?” A two-independent-groups experiment that compares beginning students in 2011 with beginning students in 2009 is comparing two cohorts of students. By contrast, a paired design experiment that compares beginning students in 2009 with the same students in 2011 is assessing change in a single cohort of students. At second testing, those students may be different not only because the world is different in 2011 but because they are older and have experienced 2 years at university. Either experiment may give interesting findings, but they are investigating different interpretations of our original question.

Choosing Target δ , Correlation, and α

Where do the values come from that I’m plugging into ESCI to calculate power? That’s a good question. We need to choose values appropriate for our particular research situation. I’ll start by discussing three approaches to choosing δ and correlation values. I’ll focus on target δ , but similar considerations apply to choosing the correlation in the paired design case.

First, we could look for an estimate of the true value of δ in the population. Previous research, ideally a meta-analysis of previous research, might provide a good estimate of δ . More generally, knowledge of the research field may suggest typical values. Perhaps you’ve already carried out similar experiments and they can suggest values to use in planning the next experiment? Some books suggest running a small pilot study, but usually such a study gives only a poor, imprecise estimate of δ , and we know that small differences in target δ can give large differences in power. The central problem is that, if we had good knowledge of δ , we’d hardly need to run our planned experiment at all.

A second approach is to choose for target δ a value that’s of theoretical or practical interest. If the theory you are testing predicts that there’s a small to medium-sized effect, you could choose $\delta = 0.3$ as your target ES. Or you might judge a decrease of 3 points on the depression scale you are using to be the minimum improvement that’s of clinical interest, or of value to the client, so you choose 3 points as your target ES. You need as well an SD estimate for the depression scale to use as a standardizer. The manual published with the scale, or previous research with the scale, may provide an SD value for an appropriate population. The discussion in Chapter 11 about finding a suitable standardizer may suggest further options. Divide 3 by your chosen SD to get a value of δ to use as the target population ES in your power calculation.

The third and most common approach is to recognize that there usually isn't just a single value of target δ that's of interest. Often it's useful to calculate power for various combinations of values relevant for our intended research. My imagined discussion between you and your colleague as you plan your HEAT experiment is an example, and illustrates how power calculations often alarm researchers about how weak their contemplated experiment is, then guide them to some compromise between what size experiment is achievable and what target δ that experiment can seek with reasonable power.

Now let's consider α . A strategy that's often recommended is to choose an α level that reflects the costs and benefits of making Type I and II errors in a particular situation. For a life-and-death decision about approving a drug for public use we might use $\alpha = .001$, which Figure 12.1 shows would give, for our earlier example, power of only .32 (the theoretical expectation for the *** column). Low power, yes, but possibly intelligent decision making if we are using NHST. On the other hand, for an exploratory experiment in a little-researched area, we may be looking for hints of anything that could on further investigation prove interesting. We might elect to use $\alpha = .10$, or some other high value, to reduce the chance of a Type II error. Power would be high, but so would the risk of Type I error. Considering costs and benefits is an appealing strategy for choosing α , but conventions about using $\alpha = .05$ or $.01$ are strong, and few researchers would consider α greater than $.05$.

Does it strike you as disturbingly subjective to have to choose so many values to calculate power? If so, I agree, and I find it even more disturbing that above I could give only vague advice about how to select target values for δ and the population correlation—two features of an experiment that are especially influential on power. Earlier I mentioned the possibility of fudging a power calculation. Yes, the need to specify so many values gives, unfortunately, large scope to choose a set of values that gives a power value to fit your needs, even if it may not represent reality. For target δ there may even be scope to choose both the numerator (3 points on the depression scale, but why not 4?) and the denominator SD we need as a standardizer. A small amount of tweaking of each and we might be able to argue that any of a wide range of δ values is reasonable. I suspect that many power analyses presented to funding bodies and ethics review boards reflect some creative adjustment of various assumptions so the final proposed sample sizes are more or less achievable and target δ values not too unbelievable. You may now be thinking that a quite different approach based on estimation and meta-analysis is especially appealing. Well, I'd never

Exploring how power varies for different values of N , α , and target δ (and, for a paired design, correlation) can give insight, but subjectivity of variable choice means power calculations are open to fudging.

discourage that thought. Next I'll turn to some ESCI pages intended to help build accurate intuitions about power.

Intuitions About Power

There are few formulas in this chapter because I'm more interested in intuitions. Also I can't give you a simple formula for tail areas under the noncentral *t* distribution, and Figures 12.3 and 12.4 show that we need such tail calculations to find power. We'll let ESCI do the number crunching while we focus on the intuitions. Much of my previous discussion was about how different aspects of an experiment have smaller or larger influences on power. This section continues the theme, but with pictures.

The Two-Independent-Groups Design

Figure 12.8 shows power curves that illustrate how power changes with *N* for the two-independent-groups design, using *t* as the test statistic. Sample size *N* is assumed the same for each group. For groups of different sizes, use **Simulate two**, as discussed in the previous section. The figure uses $\alpha = .05$, and each curve is for a particular value of δ . The curve for $\delta = 0.5$ is highlighted, and three other curves are labeled with their δ values. A

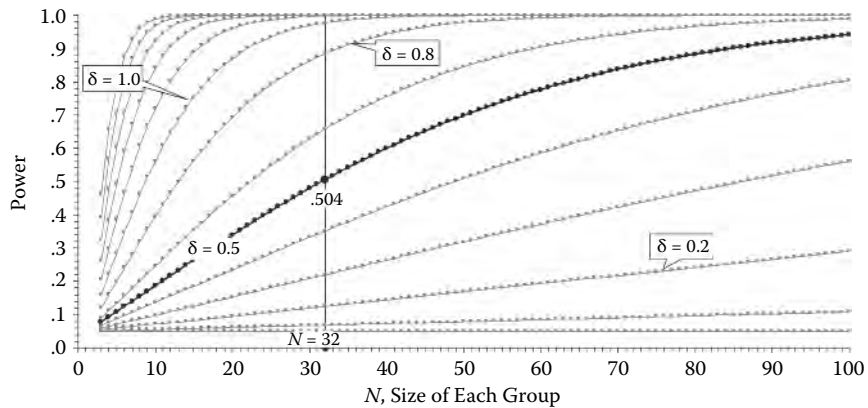


FIGURE 12.8 Power curves for the two-independent-groups design, with $\alpha = .05$, using *t* as the test statistic. Sample size *N*, the same for each group, is plotted on the horizontal axis. Each curve is for a particular value of target δ . The curve for $\delta = 0.5$ is highlighted, and three other curves are labeled with their δ values. The *N* cursor is positioned for power close to .5, for $\delta = 0.5$, and shows $N = 32$ gives that power.

vertical cursor reports power for a chosen N , in this case showing power = .50 when we have two groups of size 32.

Study Figure 12.8 for a moment and consider what messages it's sending. For me, it's saying that target δ is the most important influence on power, and that the power curve is quite different in shape for different values of δ . Even modest experiments, with two groups of 20 or 30, have high power to find large or very large population effects— δ of 0.8 or more. But if we're looking for small effects— δ of 0.2 or 0.3—even groups of 100 have very low power.

Figure 12.9 is similar, but for $\alpha = .01$. It shows in addition a horizontal cursor for power, which is set to power = .8. The vertical N cursor has been placed where the power cursor intersects the $\delta = 0.5$ curve, and shows that, with $\alpha = .01$, two independent groups of size 95 have power of .8 to find a population effect of size $\delta = 0.5$. Compare the two figures. To me they say that, for large and very large δ , power is still in most cases quite high when $\alpha = .01$. However, for small and medium effect sizes, lower α comes at the price of considerably lower power. Not even large N —well, $N = 100$, anyway—can give large power with $\alpha = .01$ unless we're looking for a substantial δ .

Figures 12.8 and 12.9 come from the **Power** two page of **ESCI** chapters 10–13. Figure 12.10 shows that page's control area. At red 1, click to assume that σ is known and z is the test statistic, or σ is not known and therefore t is the test statistic. The spinner at red 1 selects α . Be warned that, when using t , if you change α the whole figure must be recalculated, which requires roughly 3,000 noncentral t calculations. It can take a while, and

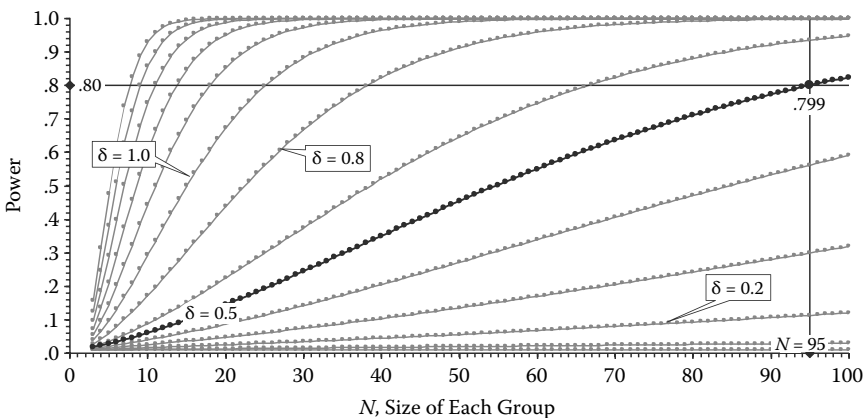


FIGURE 12.9

Same as Figure 12.8 but with $\alpha = .01$. A horizontal power cursor has been added, at power = .8. The N cursor has been moved to where the power cursor cuts the $\delta = 0.5$ power curve, and shows that $N = 95$ for that power and population ES.

1 Set up

Two-tail α

☐ Use Z, assume σ known

☒ Use t , assume σ not known

2 Effect size

☒ Label three δ series

☒ Highlight a d series

δ

☒ Label highlighted 6 series

3 Sample size

☒ Display vertical N cursor

5 Power

☐ Display horizontal power cursor

Power

6 Power for large N , using Z

N

Power

FIGURE 12.10
Control area for the **Power two** page of **ESCI chapters 10–13**. For N greater than 100, use the bottom panel. The values shown tell us that, for $\alpha = .05$ and target $\delta = 0.2$, an N of 309 is needed to achieve power of .7.

nothing seems to be happening. (Here’s a hint: If you want to change α by more than a click or two of the spinner, select z , change α , and then switch back to t .) The spinner near red 2 selects which curve is highlighted. As usual, explore the checkboxes that control various display features. When the N cursor is displayed, the slider below the figure shifts it left and right. When the power cursor is displayed, the spinner near red 5 shifts it up and down. For values of N larger than 100, which lie beyond

TABLE 12.1
Minimum Sample Size Required to Achieve Various Levels
of Power for the Two-Independent-Groups Design, for Selected
Values of α and δ

Target Power	δ	$\alpha = .05$					$\alpha = .01$				
		0.2	0.3	0.5	0.8	1.0	0.2	0.3	0.5	0.8	1.0
.50		193	87	32	14	9	332	148	55	23	15
.60		245	109	41	17	11	401	178	66	27	18
.70		309	138	51	21	14	481	214	79	32	21
.80		393	175	64	26	17	584	260	96	39	26
.90		526	234	86	34	23	744	331	120	49	32
.95		650	289	104	42	27	891	396	143	58	38

Note: Values are sample size N , where N is the size of each of the two groups. The test statistic is t except that, for $N > 100$, the test statistic is z . For such large sample sizes, using z gives the same or very similar N to that found using t .

the horizontal axis in the figure, use the bottom panel at red 6. This panel calculates power using z , but for such large N the results are very similar to those based on t .

Table 12.1 provides a further approach to building intuitions about power. It shows the smallest N needed to achieve target power shown at the left, for selected values of α and δ . The test statistic is t , except that z is used when $N > 100$.

The Paired Design

For the paired design, use the **Power paired** page of **ESCI chapters 10–13**. Figure 12.11 is from that page and shows the power curves for the paired design, for $\alpha = .01$, using t . The horizontal axis shows N , the number of pairs. The population correlation between the two measures is shown as ρ (Greek rho) and is set to .70. Figure 12.12 shows the control area, including the slider near red 1 that sets ρ . The cursors in Figure 12.11 illustrate that 40 is the smallest N that gives power at least .90. Compare Figures 12.11 and 12.9, and note that $\alpha = .01$ in both. Even with ρ only .70, a correlation likely to be exceeded in many practical cases, the considerably higher values of power in Figure 12.11 illustrate once more the advantages of the paired design—assuming, of course, that it’s appropriate for the research situation.

The **Power paired** page provides similar scope for setting values and for exploration of the paired design as **Power two** provides for the two-independent-groups design. The main addition is the slider near red 1 that sets ρ , the population correlation between the two measures. Table 12.2 gives a further perspective on power for the paired design. The left half

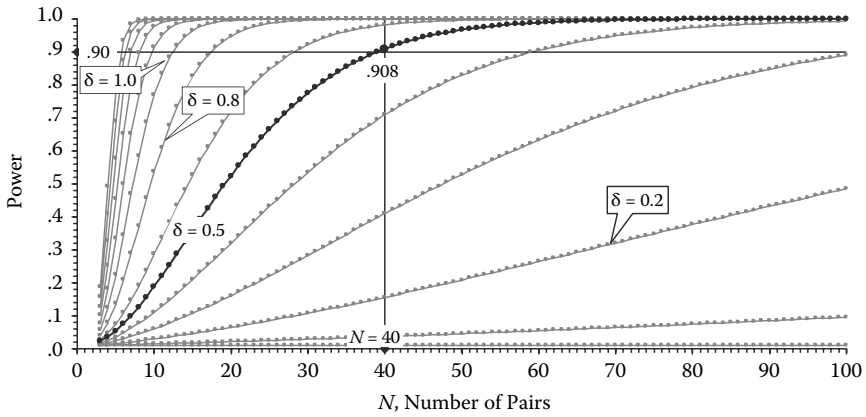


FIGURE 12.11

Power curves for the paired design, with $\alpha = .01$, test statistic t , and population correlation $\rho = .70$ between the two measures. The horizontal axis shows N , the number of data pairs. The curve for $\delta = 0.5$ is highlighted, and three other curves are labeled. The power cursor marks .90, and the N cursor shows 40 is the smallest N giving at least that power.

of the table is for $\rho = .50$, and the right half is for $\rho = .80$. Both use $\alpha = .01$, so you can compare any N in Table 12.2 with those in the right half of Table 12.1, which also uses $\alpha = .01$.

Enjoy the generally high power of the paired design, even with modest values of ρ , for example, in the range .4 to .7. In practice, the paired design often gives ρ of .8 or even .9 or more, and so even quite small experiments can have considerable power. Pretest-posttest experiments with a single group of participants are a common example of a paired design. They often have a high correlation between the pretest and posttest

A repeated measure design, such as the paired design, can give high power, but make sure the design is appropriate for the particular circumstances.

measures, and so are likely to give high power. As usual, we need to be sure that the paired design is appropriate for our situation, and we don't, for example, have serious carry-over effects between the two measures.

After all this exploration of how power varies with α , δ , N , and other features of an experiment, I need to mention, or rather warn you about, post hoc power. Then we can consider ways to increase power.

Post Hoc Power—Illegitimate Power

There's a sinister side to political power, and most people know to watch out for it. Unfortunately, there's also a sinister side to statistical

1 Set up

Two-tail α 0.01

Correlation ρ 0.8

☐ Use Z, assume σ known

☒ Use t , assume σ not known

2 Effect size

☒ Label three δ series

☒ Highlight a δ series

δ 0.2

☒ Label highlighted δ series

3 Sample size

☒ Display vertical N cursor

5 Power

☐ Display horizontal power cursor

Power

6 Power for large N , using Z

N 179

Power .951

FIGURE 12.12
Control area for the **Power paired** page of **ESCI chapters 10–13**. For N greater than 100, use the bottom panel. The values shown tell us that, for $\alpha = .01$, $\rho = .80$, and target $\delta = 0.2$, an N of 179 is needed to achieve power of .95.

power, and we all need to watch out for that. I’m referring to *post hoc power*, and I’m going to recommend that you never use it and are careful not to be misled by reports of power that are actually values of post hoc power.

The key is δ : What value of target δ is used to calculate power? I’ve discussed various ways to choose a target value for δ , but the one possibility

Post hoc power is calculated after completing the experiment, using as target δ the effect size d obtained in the experiment. It can easily mislead, so never use it.

TABLE 12.2

Minimum Sample Size Required to Achieve Various Levels of Power for the Paired Design for $\alpha = .01$, for Selected Values of Correlation ρ , and Population Effect Size δ

Target Power	δ	$\alpha = .01, \rho = .50$					$\alpha = .01, \rho = .80$				
		0.2	0.3	0.5	0.8	1.0	0.2	0.3	0.5	0.8	1.0
.50		166	78	30	14	10	70	33	14	8	6
.60		201	93	36	16	12	84	39	17	9	7
.70		241	107	42	19	13	100	47	19	10	8
.80		292	130	51	22	16	117	56	23	11	9
.90		372	166	63	27	19	149	70	28	13	10
.95		446	198	75	32	22	179	83	32	15	11

Note: Values are N , the number of pairs. The test statistic is t except that, for $N > 100$, the test statistic is z . For such large sample sizes, using z gives the same or very similar N to that found using t .

I’ve not mentioned is using the result of the experiment itself. You may have wondered, why all this concern to find a target value of δ by considering previous research, or choosing an ES likely to be of practical importance? Why not simply carry out the experiment then use d , the ES obtained by the experiment, as target δ for power? Well, yes, we can easily do that, and calculating power using our obtained d for target δ gives us what’s called *post hoc* or *observed* power.

For example, Figure 12.6 shows that, when target $\delta = 0.2$, the **Simulate two** page of **ESCI chapters 5–6** reports power of .21 for our two-independent-groups HEAT experiment, with groups of size 80 and 60 and two-tailed $\alpha = .05$. I used that page to run some simulations of the experiment. The first gave $d_{\text{unb}} = 0.29$ as the unbiased estimate of δ . Figure 12.13 shows the power panel, as in Figure 12.6, but now I’ve clicked to show also the value of post hoc power, which is .4. In other words, after running the

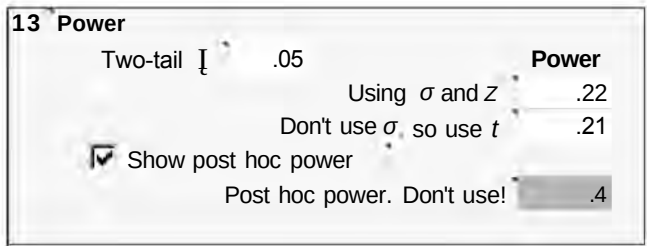


FIGURE 12.13 Part of Figure 12.6, but now post hoc power is displayed. The two-independent-groups HEAT experiment obtained $d_{\text{unb}} = .29$, and appears as Replication 1 in Table 12.3.

TABLE 12.3
Results of 10 Replications of the Two-Independent-Groups
HEAT Experiment Shown in Figure 12.6

Replication	Obtained d_{unb}	Two-Tailed p	Post Hoc Power
1	0.29	.09	.40
2	−0.07	.70	.07
3	0.20	.25	.21
4	−0.17	.32	.17
5	0.47	.007	.77
6	0.23	.18	.27
7	0.07	.69	.07
8	−0.02	.90	.05
9	0.11	.52	.10
10	0.56	.001	.90

experiment and observing $d_{\text{unb}} = 0.29$, ESCI used that d_{unb} as target δ and calculated power to be .4.

First, there’s a small logical problem with post hoc power. Power is the probability that an experiment will reject H_0 if an effect of the stated target size, δ , exists. However, after completing the experiment we’ve either rejected H_0 or we haven’t, and so it doesn’t make sense to talk about “the probability that our experiment will reject H_0 .” OK, we carefully say, “Our experiment, *before we ran it*, had power of .4 to find target $\delta = 0.29$.” That’s fine. Now let’s see what happens if we repeat the experiment a few times.

Table 12.3 shows the results of the first 10 replications of our HEAT experiment, as given by **Simulate two**. The first replication gives post hoc power as shown in Figure 12.13. Based on the values in the table, we can make statements like, “Our experiment, before we ran it, had power of .07 to find target $\delta = -0.07$ ” (that’s based on Replication 2) or, “Our experiment, before we ran it, had power of .90 to find target $\delta = 0.56$ ” (based on Replication 10).

Those are all perfectly acceptable statements, but do they tell us anything useful? Table 12.3 illustrates the close inverse relation between d_{unb} (or d) and p that we observed in Chapter 5: Small d_{unb} (or d) gives large p , and large d_{unb} (or d) gives low p . It also illustrates the very large variation of p over replication that we saw in the dance of the p values. Examine the d_{unb} and post hoc power columns: Post hoc power has a close and direct relation with d_{unb} . Now examine the p and post hoc power columns: Post hoc power has a strong inverse relation with the p values. Post hoc power, like d_{unb} and p , varies greatly with replication, but the main problem is that it doesn’t tell us anything further about our experiment beyond what d_{unb} and p have already told us. It’s a mere restatement of our experimental result. The danger is that any statement of a power value is likely to

be read as telling us something about the *experiment*, whereas a post hoc power value really only tells us about the result of running the experiment once. A post hoc power value can thus be severely misleading, especially if it's not accompanied by the full statement, as in my examples above, including the target δ value.

Hoenig and Heisey (2001) explained in more detail the reasons for avoiding post hoc power. They described a number of incorrect beliefs held by some researchers about how post hoc power might be able to assist in the interpretation of results found to be not statistically significant. I have never seen a convincing case made for any worthwhile use of post hoc power. I recommend never using it, and I express my disapproval by the gray background shading and the comments at the power panel in **Simulate two** and **Simulate paired**, where you can click if you insist on seeing post hoc power.

I've labeled post hoc power as *illegitimate power* to highlight the contrast with *legitimate power*, which uses a target δ that's chosen for its research

Legitimate power is my name for power calculated for a target δ that's chosen as being of research interest. Contrast it with post hoc power.

interest. It's legitimate power that we've discussed at length in earlier sections of this chapter. Usually we calculate it as part of planning, in advance of running the experiment, but you can calculate

legitimate power after collecting the data. The crucial point is that target δ is chosen for its research or practical interest, and is not merely the result of our experiment.

So, why am I going on at such length about something we should all avoid? I'm doing so because some widely used computer packages report post hoc power simply labeled as "power" or "observed power." No doubt many users, not understanding how legitimate and post hoc power differ, simply accept what the computer says and report that value as *the*

If statistical software reports power without specifying a population ES, it's probably post hoc power. Don't use that value. Be very cautious.

power of their experiment. No doubt it's an awkward extra step for a software package to ask the user for a target value of δ before it calculates power. However, I believe it's reprehensible for software to

duck the issue of choosing an appropriate target δ , and simply report post hoc power. At the very least, any post hoc power value should come with a clear label and warning. My recommendation is never to calculate post hoc power and always to watch out for post hoc power being passed off as legitimate power. I strongly recommend the wonderful free software G*Power 3 (tinyurl.com/gpower3). G*Power calculates power for numerous designs and measures beyond the few simple cases I discuss in this chapter and that ESCI supports.

Informativeness

Let's step aside from statistical power for a moment and consider the fundamental aim of experimental research, which is to inform us about the world. We want experiments that answer our research questions, provide useful knowledge, and give us insight into how the world works. We don't have an agreed name for this desirable feature of an experiment, but "strong," "insightful," "good," or "sensitive" come close. However, I think "informative" is best: An experiment is highly *informative* if it gives us useful information and insight about the world. Informativeness is highly desirable and, almost always, the more of it the better.

I refer to the *informativeness* of an experiment as its ability to give information to answer research questions, or to give insight about the world.

I'm introducing informativeness because I want to consider, very generally, how we can improve our experiments. Many books and articles give wonderful advice about ways to improve research. Often they focus on how to increase statistical power, but most of the advice actually applies more broadly and is valuable even if we don't use NHST. I want the benefit of the advice, but without having to use NHST or power. I'll therefore try as much as possible to interpret the advice in terms of increasing informativeness rather than statistical power. If that succeeds, we can follow the advice, but without using NHST.

Many strategies to improve experimental design are likely to increase both informativeness and statistical power. Examples include using larger N , using larger p in a paired design, and finding ways to reduce measurement error. On the other hand, increasing α will increase statistical power but is unlikely to change informativeness—the experiment is unlikely to give us greater insight if we use $\alpha = .05$ rather than $.01$. Informativeness and statistical power are related, but informativeness is the more general concept and reflects better the value of an experiment. Increasing informativeness is more generally beneficial than increasing statistical power. However, I'm not going to attempt any quantitative definition of informativeness, which will thus remain a more abstract concept, in contrast to statistical power for which we can calculate a numerical value if we make certain assumptions.

Informativeness is related to, but more fundamental than, statistical power, which has meaning only within NHST.

Back in Chapter 2 I introduced Jacob Cohen, the great statistical reformer. Box 2.2 described evidence, first presented by Cohen in 1962, that published research typically has depressingly low power. If researchers realized this, Cohen thought, surely they would stop wasting their time and insist on bigger, better-designed experiments with higher

power? His book about power (Cohen, 1969, 1988) was pioneering and remains an important resource, but, alas, Cohen's campaign was not fully successful. Researchers may now be more aware of power, but Maxwell (2004) reported that later surveys have found that the average power of published research is still, in many cases, about as low as Cohen found half a century ago.

Cohen discussed power but his argument can, I believe, be read in terms of informativeness. He wanted researchers to improve their designs and use larger sample sizes—and these are ways to increase informativeness. Maxwell (2004) also couched his advocacy of improved practices in terms of power, but in addition explained the importance of estimation and meta-analysis for achieving scientific progress. Again, I see his main argument as addressing informativeness rather than the narrower concept of statistical power. The *Publication Manual* (APA, 1994) introduced in its fourth edition the advice to “take seriously the statistical power considerations associated with your tests of hypotheses.... Routinely provide evidence that your study has sufficient power to detect effects of substantive interest” (pp. 16–17). Almost identical statements appeared in the fifth and the current sixth editions. That's NHST language, but I'd like to think that the intent is broader: to improve the informativeness of research.

I'm arguing that, if we don't use NHST, and therefore don't use statistical power, the substance of arguments in favor of higher power can usually be recast in terms of increased informativeness. Doing this may actually strengthen those arguments, for example, by removing any hint that merely increasing α , which increases power but not informativeness, might suffice. Increasing informativeness is a fundamental goal of research planning, and thinking in terms of informativeness may allow us to avoid NHST if we wish, but still benefit from so much excellent advice about experimental design.

High Power, High Informativeness

Any discussion of power should consider how power can be increased to improve the experiments being planned. However, I've argued that, more fundamentally, we should be trying to increase informativeness, not merely statistical power. I'll therefore hold until the next chapter my general discussion of how, during research planning, we can improve our intended experiments. There are just two issues I'd like to discuss here in the context of power. They both relate to the question, “Can power be too high?”

In many research situations it's difficult to achieve power that the researchers regard as sufficiently high. A number of studies combined by meta-analysis may be needed before even a moderate amount of evidence can be accumulated. By contrast, consider surveys involving many thousands of people. With such very large samples, CIs are very narrow and even tiny differences are highly statistically significant. It's a waste of time to calculate power or a p value: Power is luxuriously high whatever realistic target δ we choose, and p is minuscule. The focus needs to be on the effects themselves and their interpretation. Some may be highly statistically significant, yet so tiny that they have no importance. It's not a problem that power is too high, but thinking about power or p values may distract us from more important concerns, including the representativeness of our samples and the quality of our survey measures.

Second, note that any experiment has costs, which must be balanced against the value of the knowledge it's likely to provide. Is the size of our proposed experiment justified? Would a smaller, less costly study suffice for our research purposes? These questions are especially pertinent when costs include discomfort and risks to participants. It's highly valuable to know that a small daily dose of aspirin can reduce the risk of heart attack. However, gaining such knowledge usually requires many participants to be randomly assigned to receive the inert placebo pill, and so miss any benefit of aspirin. Clearly, such studies need to be scrutinized by an independent ethical review board before being approved. An analysis based on precision may be better, but an analysis of statistical power is one way to give guidance about the proposed study, so it's large enough to give useful knowledge, but no larger than it needs to be, to minimize participant harm.

Reporting Power

If we decide to calculate power, how should we report it? As I mentioned, the *Publication Manual* advises us to "routinely provide evidence that the study has sufficient power to detect effects of substantive interest" (APA, 2010, p. 30). However, it gives no advice on how to report power, and no power values appear anywhere in its numerous examples.

In medicine, some articles report power, but what about in other disciplines? I took a peek at *Psychological Science*—just one journal in one discipline—but I suspect that what I found is typical for many journals, at least across the social and behavioral sciences. I conducted a full-text search for "power" in the 259 articles published in *Psychological Science* in the last year.

Just 22 articles (8%) mentioned statistical power, of which four reported power values, ranging from .79 to .99. Another 10 made brief comments about experimental design or statistical analysis, mentioning, for example, the combination of groups or the use of meta-analysis as strategies to increase power. All those comments were reasonable, and all can be read as referring to informativeness.

The remaining eight articles all referred to power while discussing effects that were not statistically significant. For example, they suggested that considering power may justify accepting the null hypothesis, or that a result that was found not statistically significant might, with higher power, achieve statistical significance. However, even high power rarely, if ever, provides good grounds for accepting a null hypothesis and concluding that we have a zero effect. The CI always provides much clearer guidance for interpretation, as we've seen in earlier chapters.

The articles did not provide a good example of how to report power—if you must. Suppose you wish to present one of our HEAT experiments to your ethics review board, which requires a power analysis. In the context of your research goals, and bearing in mind available participants and resources, you need to justify your choices of experimental design, α , target δ , ρ (if a paired design), and N . It may be useful to describe how power changes for different values of those variables before you justify the set of values for which you seek approval. Pay particular attention to explaining your choice of target δ . Be as realistic as you can, bearing in mind the risk of fudging. In a journal article a briefer version is needed, but allocate a few words to justifying your choice of target δ .

However, I have yet to find a case in which I think a power analysis is more illuminating than a precision analysis could be. I hope funding bodies and ethics review boards will accept a precision analysis instead and, soon, require one. Considering journal publication, very few researchers have “routinely provide[d] evidence that the study has sufficient power” (APA, 2010, p. 30), as the *Publication Manual* requires. I see no signs that researchers are likely to adopt this advice. It's one of the few guidelines in the *Manual* I don't endorse.

It's almost time to embrace the new statistics and move beyond NHST and statistical power to *precision* as a tool for planning research. That's the topic of the next chapter.

It's time for take-home messages. To help you write yours, you may want to look back over the figures in this chapter. We started with a simulation and a wide spread of p values, then came the power picture for the single-group design, using noncentral t . Two pages in **ESCI chapters 5–6** allow you to calculate power for the two-independent-groups and paired designs. How should we select target δ ? Two pages in **ESCI chapters 10–13** present power curves for the two-independent-groups and paired

designs. Intuitions about power are important, especially understanding which factors influence power most strongly. Avoid post hoc power! Consider the informativeness of an experiment, not just statistical power. How should we report power, if we choose to?

Exercises

- 12.1 You are planning a study of attitudes to the length of jail sentences for homicide, using a scale running from -4 to $+4$, where 0 indicates a judgment that current sentences are about right. Previous research suggests that the population SD for the scale is 1.2 . You plan to use a single sample and would like to be able to detect a true effect of 0.5 scale units, using $\alpha = .01$. If you use $N = 100$, what is the power? Explore power for various other choices of effect size, α , and N .
- 12.2 You wish to use the same scale to compare attitudes in two very different neighborhoods. You would like to be able to detect a difference of 0.3 scale units. Consider power and make recommendations.
- 12.3 Again using the same scale, you wish to compare attitudes in a single neighborhood, using a single sample of people who respond both before and 3 months after a state election. You would like to be able to detect a change of 0.2 scale units. Previous research using the scale has reported correlations between pre-test and posttest attitudes in the range $.72$ to $.89$. Consider power and make recommendations.
- 12.4 If you are using another statistics textbook, find worked examples or exercises about power for which the ESCI pages discussed in this chapter are helpful. Compare the methods used by that other book with ESCI's, and also compare the results.
- 12.5 A leading researcher in clinical psychology uses the rule of thumb that, roughly speaking, most well-established types of psychotherapy give an improvement of around $\delta = 0.5$ to 0.7 , whereas comparisons between any two such therapies are likely to give differences of around $\delta = 0$ to 0.3 . Suppose you are planning a two-independent-groups experiment to assess such a therapy, using target $\delta = 0.6$, and a second two-independent-groups experiment to compare two such therapies with target $\delta = 0.2$ for the difference. Compare the two experiments you will need.

- 12.6 What does the tiny invisible area to the left under the H_1 curve in Figures 12.3 and 12.4 represent? Why is it so small?
- 12.7 Invent a game to help build your intuitions about power. Challenge a friend to estimate power for an experiment you describe—you'll need to state the design, and also α , δ , and N (and ρ if a paired design). Practice using the **Power two** and **Power paired** pages of **ESCI chapters 10–13** until you can find answers quickly. Develop the game to focus on estimating just one factor, while holding others constant. For example, for a two-independent-groups experiment with $\delta = 0.5$ and $\alpha = .01$, what is the power for $N = 100$? What is N to obtain power = .80? How big a δ can you detect if $N = 100$ and power = .80?
- 12.8 Use that game, or relevant ESCI pages, or Tables 12.1 and 12.2 to identify some sets of values that seem especially relevant for your discipline. Choose a handful of these to remember, to serve as benchmarks for judgments about published research you read.
- 12.9 Explore any statistical software you use. Does it give you a value for power? Is this legitimate power, or post hoc power? Is it clearly labeled?
- 12.10 Even if that software reports post hoc power, perhaps as a default, try to find a way to have it calculate and report legitimate power, after you specify a target population effect size.
- 12.11 Download G*Power 3 (tinyurl.com/gpower3). Explore as you wish. Use it to check one or two values in Table 12.1 or 12.2.
- 12.12 Find a journal article in your discipline that mentions statistical power. If possible, find an article that reports a power calculation. Do these articles discuss and use power appropriately? Can you make any recommendations? You could try a full-text search for "power," as I used to investigate power in *Psychological Science*. You should be able to conduct a single search for a whole selected date range of journal issues, before opening only those articles identified as containing the word "power."
- 12.13 Find a journal article in your discipline that could usefully mention statistical power or report a power calculation, but doesn't. Make your recommendations.
- 12.14 Larger N gives smaller variation in M over replication—in other words, the dance of the means is narrower for larger N . Similarly, s varies less over replication when N is large. In Chapter 5 we found, perhaps surprisingly, that for a given initial p value, the variation in replication p does not change with N , even for

large N . Consider the amount of variation in post hoc power over replication. For a given level of legitimate power, would you guess that variability in post hoc power decreases with larger N , or remains about the same? Use the **Simulate two** page of **ESCI chapters 5–6** to check out the variation in post hoc power. You could, for example, set up an experiment with $N_1 = N_2 = 8$, then adjust δ until power is about .7. Take 20 experiments and write down the values of post hoc power. Then set up an experiment with $N_1 = N_2 = 64$ and a much smaller δ so power is again about .7. Again record 20 values of post hoc power. Is the amount of variability similar? Try other experiments if you like. What do you conclude about post hoc power?

- 12.15 Consider every way you can think of to increase statistical power. For each, decide whether you are also increasing the informativeness of the experiment.
- 12.16 Revisit your take-home messages. Improve them and extend the list if you can.



Take-Home Messages

- I'm ambivalent about statistical power, which is only defined in the context of NHST, so the new statistics do not use it. We should keep in mind our new statistics intuitions even while discussing power within NHST.
- Statistical power is the probability of rejecting H_0 if there is a true population effect of size target δ . The value of power depends on the experimental design, N , α , target δ , and in the paired case ρ , the population correlation between the two measures.
- The informativeness of an experiment is its ability to answer our research questions and give us information about the world. Informativeness is more fundamental than statistical power.
- It's often difficult to achieve high power, and the power of published research is often disappointingly low, despite the efforts of Jacob Cohen and other statistical reformers.
- The power picture shows the distribution of the test statistic when H_0 is true and when H_1 is true. For σ known the test statistic is z and the two distributions are normal. For σ not known the test statistic is t and the distributions are the central and noncentral t distributions. *Take-home picture:* The power picture, for example, Figure 12.3.
- The **Simulate two** page of **ESCI chapters 5–6** allows you to calculate power for the two-independent-groups design, and the **Simulate paired** page for the paired design.
- It's important to build intuitions about power. In general, larger N gives higher power and smaller α lower power, but target δ is even more influential: Small changes in δ can give substantial changes in power. The paired design can give high power, especially if ρ is at least moderately large.
- The power curves of the **Power two** and **Power paired** pages of **ESCI chapters 10–13** illustrate how power varies with N and other aspects of an experiment. These pages also give the values of N shown by Tables 12.1 and 12.2 for selected situations.
- Post hoc or observed power is calculated after data collection, using the Cohen's d obtained in the experiment as target δ . It reflects the result rather than any basic feature of the experiment, and varies greatly with replication. Don't use post hoc power, which can be highly misleading. Watch out for software that reports post hoc power without saying so.

- If you intend to use NHST, a power calculation may be useful, for example, to justify the design and sample sizes of a planned experiment.
- To adopt the new statistics and move beyond NHST, use precision rather than power.

